Advanced Modern Engineering Mathematics

fourth edition

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1 Matrix Analysis

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] Today & matlab

What is a matrix?

an array of numbers

or

a rectangular arrangements of numbers

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 5 & 6 & 7 \end{bmatrix}, \quad B = \begin{bmatrix} 1 & 1 \\ 5 & 5 \\ 7 & 7 \end{bmatrix}$$

Lets practice creating matrix in Matlab.

How about adding two matrices in Matlab?

Find determinant of matrix in Matlab. (matrix must be square)

$$A = \begin{bmatrix} 1 & 1 \\ 2 & 5 \end{bmatrix}$$

$$\text{determinant}(A) = 1.5 - 2.1$$
$$= 5 - 2$$
$$= 3$$

A is an $n \times n$ matrix

$$A = \begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \cdot & - & - & - \\ a_{n1} & a_{n2} & \cdots & a_{nn} \end{bmatrix}$$

$$A = \begin{bmatrix} 1 & 0 & 1 \\ 5 & 6 & 7 \\ 0 & 2 & 1 \end{bmatrix}$$

find determinant of A

$$\det(A) = 1 \begin{vmatrix} 6 & 7 \\ 2 & 1 \end{vmatrix} - 0 \begin{vmatrix} 5 & 7 \\ 0 & 1 \end{vmatrix} + 1 \begin{vmatrix} 5 & 6 \\ 0 & 2 \end{vmatrix}$$

$$= 1 * (-8) - 0 + 1 (10 \qquad)$$

$$= 2$$

Find inverse of a matrix on Matlab

whats a unit or identity matrix

$$I = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

Inverse: of a square matrix A :
is denoted by A^{-1}, satisfies the rule

$$A(A^{-1}) = I = A^{-1}(A)$$

Matrix multiplication in Matlab

Matrix transpose in Matlab

Last time:

→ matrix definition

→ inverse, determinant, etc.

matlab practice

Today: Solving equations → shortcut way

→ elimination method

rank of a matrix

[Vector Space]

Linear equations

$$x + y + z = 6,$$

$$x + 2y + 3z = 14,$$

$$x + 4y + 9z = 36$$

Solve this in matlab now.

$$\underbrace{\begin{bmatrix} 1 & 1 & 1 \\ 1 & 2 & 3 \\ 1 & 4 & 9 \end{bmatrix}}_{A} \underbrace{\begin{bmatrix} x \\ y \\ z \end{bmatrix}}_{v} = \underbrace{\begin{bmatrix} 6 \\ 14 \\ 36 \end{bmatrix}}_{b}$$

$$Av = b.$$

A, b : known

v : to be calculated

$$A^{-1}Av = A^{-1}b$$

$$Iv = A^{-1}b$$

$$\boxed{v = A^{-1}b}$$

Q. Can we have solution of

$$Ax = b$$ even if A is not invertible ??

$A : 1\times 1$

$b = 1\times 1$

$0 \cdot x = 1 \rightarrow$ solution does not exist

$0 \cdot x = 0 \rightarrow$ Solution exist

$$\underbrace{\begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}}_{A:} \begin{bmatrix} \\ \end{bmatrix} = \begin{bmatrix} 5 \\ 5 \end{bmatrix}$$

A: is not invertible, but solution exist

if A is not invertible, we need to fall back to elimination method.

$$\begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 5 \\ 5 \end{bmatrix}$$

Start with augmented matrix

$$\left[\begin{array}{cc|c} 1 & 1 & 5 \\ 1 & 1 & 5 \end{array}\right]$$

do row operations till lower triangular elements become zero

$$\left[\begin{array}{cc|c} 1 & 1 & 5 \\ 1 & 1 & 5 \end{array}\right] \xrightarrow{R_2 \to R_2 - R_1} \left[\begin{array}{cc|c} 1 & 1 & 5 \\ 0 & 0 & 0 \end{array}\right]$$

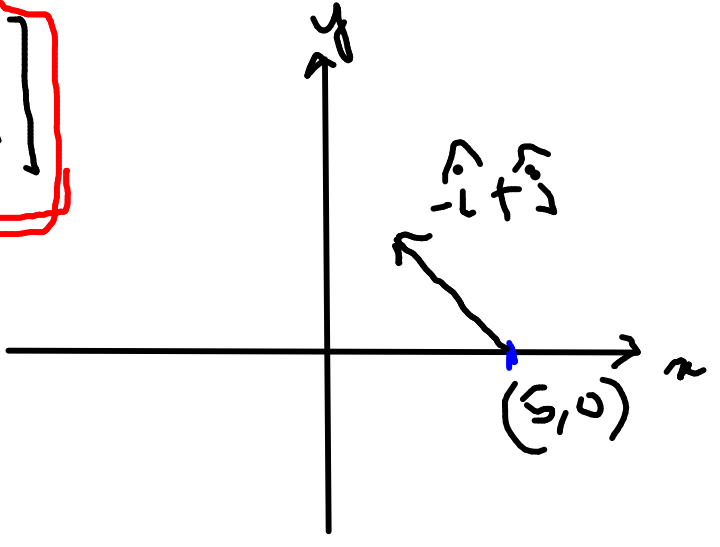
infinitely many solutions exist

$x+y=5$
$x+y=5$

$x+y = 5$
$x = 5-y$

solutions : $\boxed{\begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 5-y \\ y \end{bmatrix} = \begin{bmatrix} 5 \\ 0 \end{bmatrix} + y \begin{bmatrix} -1 \\ 1 \end{bmatrix}}$

$-\infty < y < \infty$



Rank of a matrix

Solve the 3 x 3 equation from using Gaussian elimination.

1.3 Vector spaces

Definition

A **real vector space** V is a set of objects called **vectors** together with rules for addition and multiplication by real numbers. For any three vectors $\boldsymbol{a}$, $\boldsymbol{b}$ and $\boldsymbol{c}$ in V and any real numbers α and β the sum $\boldsymbol{a} + \boldsymbol{b}$ and the product $\alpha\boldsymbol{a}$ also belong to V and satisfy the following axioms:

(a) $\boldsymbol{a} + \boldsymbol{b} = \boldsymbol{b} + \boldsymbol{a}$

(b) $\boldsymbol{a} + (\boldsymbol{b} + \boldsymbol{c}) = (\boldsymbol{a} + \boldsymbol{b}) + \boldsymbol{c}$

(c) there exists a zero vector $\mathbf{0}$ such that

$$\boldsymbol{a} + \mathbf{0} = \boldsymbol{a}$$

(d) for each $\boldsymbol{a}$ in V there is an element $-\boldsymbol{a}$ in V such that

$$\boldsymbol{a} + (-\boldsymbol{a}) = \mathbf{0}$$

(e) $\alpha(\boldsymbol{a} + \boldsymbol{b}) = \alpha\boldsymbol{a} + \alpha\boldsymbol{b}$

(f) $(\alpha + \beta)\boldsymbol{a} = \alpha\boldsymbol{a} + \beta\boldsymbol{a}$

(g) $(\alpha\beta)\boldsymbol{a} = \alpha(\beta\boldsymbol{a})$

(h) $1\boldsymbol{a} = \boldsymbol{a}$

Row reduced form

$$\begin{array}{c} \begin{array}{cccc} x_1 & x_2 & x_3 & x_4 \end{array} \\ \left[\begin{array}{cccc:c} \boxed{3} & 2 & 2 & -5 & 8 \\ 0 & \boxed{1.1} & 1.1 & -4.4 & 1.1 \\ 0 & 0 & 0 & 0 & 0 \end{array}\right] \end{array}$$

A $\qquad$ b

Pivot variables: x_1, x_2

free variables: x_3, x_4

$$1.1x_2 + 1.1x_3 - 4.4x_4 = 1.1$$

$$x_2 = 1 - x_3 + 4x_4$$

$$3x_1 + 2x_2 + 2x_3 - 5x_4 = 8$$

$$x_1 = (8 - 2x_2 - 2x_3 + 5x_4)\frac{1}{3}$$

$$= (8 - 2(1 - x_3 + 4x_4) - 2x_3 + 5x_4)\frac{1}{3}$$

Solution :

$$\begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} = \begin{bmatrix} \left(8 - 2\left(1 - x_3 + 4x_4\right) - 2x_3 + 5x_4\right)\frac{1}{3} \\ 1 - x_3 + 4x_4 \\ x_3 \\ x_4 \end{bmatrix}$$

where x_3 & x_4 are free parameters